GRIN1 (GluN1 / NR1): Functional Annotation Report

Gene: GRIN1 (HGNC:4584) · **Protein:** Glutamate receptor ionotropic, NMDA 1 (GluN1 / NR1 / NMD-R1 / zeta-1) · **UniProt:** Q05586 · **Organism:** *Homo sapiens* (Human) · **Family:** Glutamate-gated ion channel (TC 1.A.10.1)

Summary

GRIN1 encodes GluN1 (also called NR1 or NMDAR1), the obligatory, non-redundant core subunit of the N-methyl-D-aspartate (NMDA) subtype of ionotropic glutamate receptor. Its primary molecular function is not enzymatic but **ligand- and voltage-gated cation channel activity**. GluN1 contributes two interlocking capabilities to the receptor: (1) it supplies the binding site for the co-agonists **glycine and D-serine**, and (2) it co-assembles with glutamate-binding GluN2 subunits (or, alternatively, with GluN3 subunits) to build the Ca^{2+}-permeable cation channel that mediates the slow, Ca^{2+}-signaling component of excitatory synaptic transmission in the central nervous system. The canonical receptor is a heterotetramer of two GluN1 and two GluN2 subunits arranged in a **1-2-1-2** configuration. Because glycine binds GluN1 and glutamate binds GluN2, and because channel opening additionally requires membrane depolarization to relieve a voltage-dependent Mg^{2+} block of the pore, the NMDA receptor behaves as a **Hebbian coincidence detector** — opening only when presynaptic glutamate release and postsynaptic depolarization coincide. The resulting Ca^{2+} influx is the trigger for long-term potentiation (LTP), long-term depression (LTD), learning, memory, and activity-dependent neurodevelopment.

GluN1 is remarkably versatile. Beyond the classic GluN1/GluN2 NMDA receptor, GluN1 partners with GluN3 subunits to build **excitatory glycine receptors (eGlyRs)** that are gated by glycine alone, independent of glutamate. GluN1's N-terminal exon-5 splice cassette and C-terminal C1/C2 cassettes act as regulatory modules: the exon-5 insert tunes proton and neurosteroid allosteric modulation, while the C-terminal tail carries an endoplasmic-reticulum (ER) retention signal that gates surface trafficking and binds calmodulin to mediate Ca^{2+}/calmodulin-dependent inactivation — a negative-feedback brake on Ca^{2+} influx. Critically, the *same* GluN1-containing receptor produces diametrically opposite cellular outcomes depending

on *where* it sits: **synaptic** NMDA receptors drive pro-survival CREB signaling and plasticity, whereas **extrasynaptic** NMDA receptors drive excitotoxic cell death implicated in stroke and neurodegeneration.

The subunit's essential, non-redundant status is bracketed by genetics at both extremes. Complete genetic ablation of *Grin1* in mice abolishes all NMDA-evoked currents and Ca^{2+} influx and is neonatally lethal, while de novo human *GRIN1* variants cause a severe neurodevelopmental encephalopathy (GRIN1-related disorder) with intellectual disability, hypotonia, movement disorder, and epilepsy. Structural biology — crystal structures of the GluN1 S1S2 ligand-binding core and of the intact receptor — confirms the clamshell glycine-binding mechanism and the layered, Ca^{2+}-permeable pore architecture, and evolutionary/bioinformatic analyses place GluN1 within the larger ionotropic glutamate receptor family, sharing a periplasmic-binding-protein-like ligand-binding fold with the AMPA receptor GluR2. **Gene identity is confirmed and unambiguous:** the symbol, protein description, human organism, family (TC 1.A.10.1), and domain architecture all match the primary literature.

Key Findings

Finding 1 — GluN1 is the obligatory, non-redundant subunit required for all NMDA receptor function

GluN1 is not merely one subunit among several; it is the indispensable core without which no functional NMDA receptor can form. The definitive evidence comes from in vivo genetic ablation in mice. Targeted disruption of the *Grin1* (NR1) gene abolishes the NMDA response entirely: neurons from homozygous null mutants show complete loss of NMDA-inducible increases in intracellular calcium and of NMDA-evoked membrane currents, and the animals die 8–15 hours after birth from respiratory failure (PMID: 8060614). As the authors state, *"NR1 has a unique role, which cannot be substituted by any other subunit, in determining the activity of the endogenous NMDA receptor,"* and *"NMDA-inducible increases in intracellular calcium and membrane currents were abolished in neurons from homozygous null mutants."* GluN1 has no known functional homolog and is expressed in almost every neuronal cell type, underscoring its universal role in NMDA signaling (PMID: 19794189): *"The NMDA receptor (NMDAR) subunit GluN1 is an obligatory component of NMDARs without a known functional homolog and is*

expressed in almost every neuronal cell type." This obligatory, ubiquitous status is the central fact of GluN1 biology — every finding below concerns receptors that necessarily contain GluN1.

Finding 2 — GluN1 harbors the glycine/D-serine co-agonist site; glutamate binds GluN2

The division of labor within the NMDA receptor is precise and structurally defined. NMDA receptors require *both* glycine (or D-serine) and glutamate to open, with the two agonists binding distinct subunits: **GluN1 forms the glycine site and GluN2 forms the glutamate site** (PMID: 12805203): *"NMDA receptors require both glycine and glutamate for activation with NR1 and NR2 forming glycine and glutamate sites, respectively."* This assignment is proven at atomic resolution: co-crystal structures of the GluN1 (NR1) S1S2 ligand-binding core were solved in complex with the agonists glycine and D-serine, directly demonstrating that GluN1 binds these co-agonists (*"we describe the cocrystal structures of the NR1 S1S2 ligand-binding core with the agonists glycine and D-serine"*). The functional building block is the heterodimer: the **NR1–NR2A (GluN1–GluN2A) heterodimer is the functional unit** within the tetrameric receptor (PMID: 16281028): *"the NR1-NR2A heterodimer is the functional unit in tetrameric NMDA receptors."* Thus GluN1's primary biochemical function is to serve as the glycine/D-serine co-agonist sensor of the receptor complex. The physiological co-agonist supply is set by serine racemase (producing D-serine) and by glycine transporter GlyT1, which regulates ambient glycine at the GluN1 site (PMID: 37833512, PMID: 39059742).

Finding 3 — The receptor is a Ca^{2+}-permeable coincidence detector with 1-2-1-2 architecture

The intact receptor's architecture explains its signature physiology. X-ray crystal structures of the complete GluN1–GluN2B receptor reveal subunits arranged in a **1-2-1-2** fashion, with a layered organization of amino-terminal domains (ATD), ligand-binding domains (LBD), and transmembrane domains (TMD), and a pore lined by residues that coordinate Mg^{2+} and channel blockers (PMID: 25008524): *"Receptor subunits are arranged in a 1-2-1-2 fashion."* Functionally, this makes the NMDA receptor a Hebbian-like coincidence detector: *"NMDA receptors are Hebbian-like coincidence detectors, requiring binding of glycine and glutamate in combination with the relief of voltage-dependent magnesium block to open an ion conductive pore across the membrane bilayer."* The convergence of three requirements — glycine on GluN1,

glutamate on GluN2, and depolarization to expel Mg^{2+} — means the channel opens only when pre- and postsynaptic activity coincide, and the ensuing Ca^{2+} influx is the biochemical substrate for synaptic plasticity.

Finding 4 — GluN1 surface trafficking is gated by an ER retention signal in its C1 cassette

Where and when GluN1-containing receptors reach the cell surface is tightly controlled by signals within GluN1's own C-terminus. A CD8–NR1 C-terminal fusion protein is retained intracellularly in both HEK293 cells and neurons because of an **ER retention signal located in the C1 cassette** (PMID: 11640925): *"the NR1 fusion protein was retained intracellularly in both HEK293 cells and neurons because of the presence of an ER retention signal in the C1 cassette."* This retention is not absolute — it can be overridden *"either by the addition of a PDZ domain-binding motif or by mimicking phosphorylation at a site adjacent to the retention signal,"* providing regulated control of surface expression. Beyond the retention signal itself, NMDA receptors are sorted to dendrites through a specialized secretory route: NR1/NR2B complexes (with KIF17, CASK, SAP97) are *"diverted from the somatic endoplasmic reticulum into a specialized endoplasmic reticulum subcompartment that bypasses somatic Golgi"* and use dendritic Golgi outposts, a pathway distinct from that used by AMPA receptors (PMID: 19620977). Additional ER-retention determinants in the M3 transmembrane segments ensure that only properly assembled complexes reach the surface (PMID: 18836292). Trafficking regulation is therefore an intrinsic property of the GluN1 subunit.

Finding 5 — De novo GRIN1 variants cause a severe NMDAR channelopathy

Human genetics confirms the subunit's essential role from the disease side. De novo *GRIN1* mutations cause a severe early-infantile neurodevelopmental encephalopathy characterized by intellectual disability, hypotonia, hyperkinetic and stereotyped movements, and epilepsy (PMID: 28389307): *"De novo GRIN1 mutations have recently been shown to cause severe intellectual disability, hypotonia, hyperkinetic and stereotyped movements, and epilepsy."* Mechanistic characterization links genotype to channel dysfunction: a recurrent de novo variant, **GluN1-M641I**, produces gain-of-function channels showing *"enhanced agonist potency and reduced Mg^{2+}"* block (PMID: 34227748). Variants in the pre-M1 / S1-M1 linker region of GRIN1 (and the paralogous GRIN2 genes) critically alter channel gating, agonist potency, open probability, and surface expression; this helix is essentially devoid of variation in healthy populations while being a hotspot for pathogenic de novo variants (PMID: 37000222). A

conserved M4 glycine acts as a gating hinge where disease-associated mutations produce slow deactivation and high Ca^{2+} flux (PMID: 30217972). These disease mechanisms directly corroborate GluN1's role in co-agonist binding, Mg^{2+} block, and gating.

Finding 6 — GluN1 partners with GluN3 to form glycine-gated excitatory receptors (eGlyR)

GluN1's functional repertoire extends beyond classic NMDA receptors. When assembled with GluN3 subunits rather than GluN2, GluN1 forms receptors with fundamentally different pharmacology: unlike GluN2-containing NMDARs that require both glutamate and glycine, **GluN3-containing receptors are activated solely by glycine** (PMID: 42665662): *"Unlike GluN2-containing NMDARs, GluN3-containing receptors are activated solely by glycine."* Native GluN3A-containing receptors are *"diheteromeric assemblies"* of GluN1 and GluN3A, and cryo-EM shows glycine opens the channel via conformational changes in GluN3A. These **excitatory glycine receptors (eGlyRs)** have a defined physiological role: discrete interneuron subsets use GluN1/GluN3A eGlyRs to regulate hippocampal network activity throughout development and evolution (PMID: 41472680). Thus GluN1 is the common glycine-binding subunit shared between two functionally distinct receptor classes — the glutamate+glycine-gated GluN1/GluN2 NMDAR and the glycine-only-gated GluN1/GluN3 eGlyR. The same glycine-clamping module is deployed either as a co-agonist gate (with GluN2) or as the primary agonist gate (with GluN3).

Finding 7 — Gating requires two glycine sites; the exon-5 cassette tunes allosteric modulation

Because the receptor contains two GluN1 subunits, it contains two glycine sites, and both must be occupied for the channel to open. Channel gating requires a **quadruply-liganded state** — binding of two glutamate and two glycine molecules (PMID: 32070387): *"The quadruply-liganded state of binding of two glutamate and two glycine molecules to the receptor drives channel gating,"* enabling cation and Ca^{2+} flux. GluN1's N-terminal **exon-5 splice cassette** acts as an allosteric tuning module: the exon-5 insert of NR1-1b *"regulates the magnitude of proton-dependent PS [pregnenolone sulfate] potentiation"* and modulates tonic proton inhibition and polyamine sensitivity (PMID: 21883211). Alternative splicing of GluN1 therefore directly controls how the receptor responds to protons (pH) and neurosteroids, adding a layer of context-dependent regulation on top of the core gating machinery.

Finding 8 — The GluN1 C-terminal tail mediates Ca^{2+}/calmodulin-dependent inactivation

GluN1 is not a passive pore — its cytoplasmic tail actively limits its own Ca^{2+} throughput. Ca^{2+} influx through NMDA channels is curtailed by a rapid **Ca^{2+}/calmodulin (CaM)-dependent inactivation** that results from competitive displacement of cytoskeleton-binding proteins from the GluN1 (NR1) subunit by Ca^{2+}/CaM (PMID: 10844014): *"The influx of Ca(2+) through NMDA channels is limited by a rapid and Ca(2+)/calmodulin (CaM)-dependent inactivation that results from a competitive displacement of cytoskeleton-binding proteins from the NR1 subunit of the receptor by Ca(2+)/CaM."* The same study shows PKC phosphorylates the NR1 C-terminus, and PKC and CaM converge on steady-state current — i.e., GluN1's tail integrates kinase and Ca^{2+}-sensor signals. Calmodulin binds a high-affinity site in the GluN1 **C1 exon cassette** and a lower-affinity site in the C-terminal region (PMID: 9089692): *"NMDA receptor function can be regulated by direct binding of calmodulin to a low and high affinity (C1 exon cassette) site in the C-terminal region of the NR1 subunit."* This makes the GluN1 tail a built-in Ca^{2+} sensor providing negative feedback on channel activity — a safeguard against excessive Ca^{2+} loading.

Finding 9 — Subcellular localization dictates opposite outcomes: synaptic survival vs. extrasynaptic death

The downstream consequence of GluN1-receptor activation depends critically on *where* the receptor is located. The cellular outcome of NMDA receptor stimulation is dictated by subcellular localization: **synaptic NMDARs promote plasticity and survival**, driving CREB activation and neuroprotective gene expression, **whereas extrasynaptic NMDARs mediate excitotoxicity** and contribute to cell death in neurodegenerative diseases (PMID: 31924235): *"Synaptic NMDARs promote plasticity and survival whereas extrasynaptic NMDARs mediate excitotoxicity and contribute to cell death in neurodegenerative diseases."* This dichotomy holds in human neurons: in iPSC-derived brain organoids, *"glutamate-induced excitotoxic cell death is mediated primarily by extrasynaptic N-methyl-d-aspartate receptors (NMDARs),"* while synaptic NMDAR activity drives sustained CREB phosphorylation and neuroprotection (PMID: 34958875). The same molecular machine — a GluN1-containing receptor — thus serves as both a pro-survival plasticity engine and a pro-death excitotoxicity trigger, with location as the decisive variable. This principle underlies NMDAR involvement in ischemic stroke and neurodegeneration (PMID: 40490524).

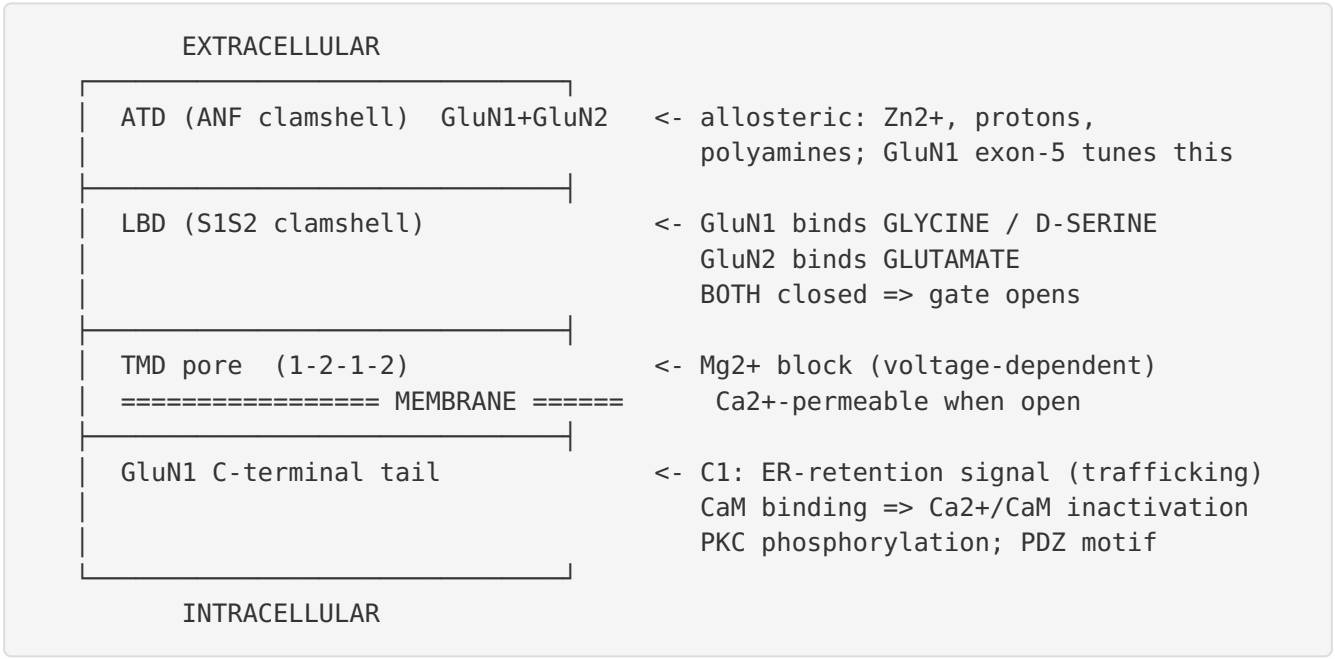
Finding 10 — GluN1's function is corroborated by its modular, evolutionarily derived domain architecture

Structure and evolution independently validate the functional assignments above. The GluN1 S1S2 ligand-binding core adopts a bilobed **"clamshell"** fold, characteristic of bacterial periplasmic binding proteins, that closes around glycine/D-serine and opens in the presence of antagonist (PMID: 12805203): *"The cleft of the S1S2 'clamshell' is open in the presence of the antagonist DCKA and closed in the glycine, DS and DCS complexes."* This domain-closure mechanism is the structural basis of gating. GluN1 is evolutionarily related to the AMPA-receptor GluR2 subunit (~40% homology in the ligand-binding core), and the partial-agonist mechanism it uses is a refinement of the paradigm established for GluR2 (PMID: 15996549): *"the NR1 subunit provides a new paradigm for partial agonist action that is distinct from that of the evolutionarily related GluR2, AMPA-sensitive receptor."* Evolutionary trace and molecular-dynamics analyses of ionotropic glutamate receptor sequences map the ligand-binding residues and show that disruption of domain closure impairs channel activation (PMID: 15597199). The InterPro domain annotations supplied with the target (ATD ANF clamshell IPR001828; Glu/Gly-binding IPR019594; ionotropic pore module IPR001508/IPR015683/IPR001320; family TC 1.A.10.1) correspond precisely to this architecture.

Mechanistic Model / Interpretation

The findings converge on a single coherent model of GluN1 as a multifunctional, obligatory receptor subunit whose domains each carry out a distinct part of the receptor's job.

Receptor architecture and the coincidence-detection logic



Opening the channel requires three simultaneous conditions (coincidence detection):

Requirement	Subunit / site	Evidence
Glycine / D-serine bound	GluN1 LBD (×2)	PMID: 12805203
Glutamate bound	GluN2 LBD (×2)	PMID: 12805203
Mg^{2+} block relieved	TMD pore (depolarization)	PMID: 25008524
→ Net: quadruply-liganded + depolarized	whole receptor	PMID: 32070387

Only when all three are met does the pore open to Ca^{2+} and other cations, converting coincident synaptic activity into a Ca^{2+} signal that drives LTP, LTD, learning, and memory.

GluN1 as a combinatorial hub

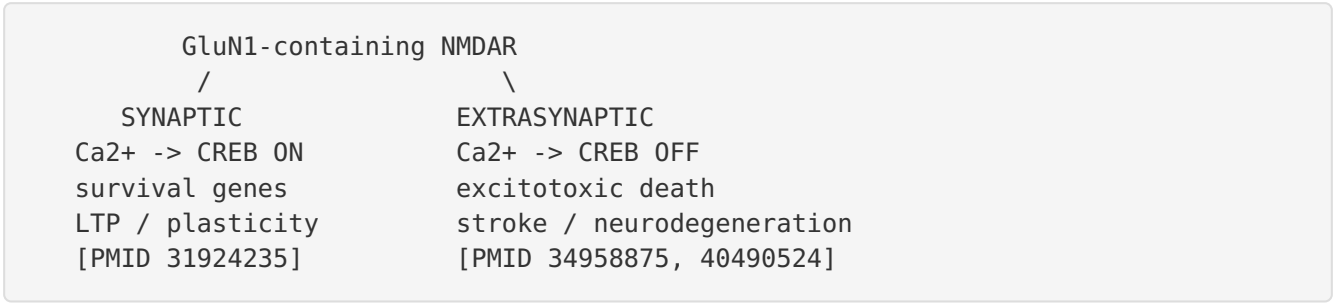
Partner subunit	Receptor type	Gating agonist(s)	Role	Evidence
GluN2 (A–D)	Classic NMDAR	Glutamate **and** glycine	Coincidence detector; plasticity, Ca^{2+} signaling	PMID: 12805203, PMID: 16281028
GluN3 (A/B)	Excitatory glycine receptor (eGlyR)	Glycine **only**	Interneuron network regulation, development	PMID: 42665662, PMID: 41472680

In both cases GluN1 contributes the glycine-binding clamshell; the identity of the partner subunit redefines whether glutamate is also required.

Regulatory layers intrinsic to GluN1

1. **Splicing (N-terminal exon-5):** tunes proton inhibition, polyamine, and neurosteroid (pregnenolone sulfate) modulation (PMID: 21883211).
2. **Trafficking (C-terminal C1):** ER retention signal controls surface delivery; overridden by PDZ motif or phosphorylation; specialized dendritic secretory pathway (PMID: 11640925, PMID: 19620977).
3. **Feedback inactivation (C-terminal tail):** Ca^{2+}/calmodulin displaces cytoskeletal anchors to inactivate the channel — a self-limiting brake on Ca^{2+} influx; PKC phosphorylation modulates this (PMID: 10844014, PMID: 9089692).

Location determines life-or-death signaling



The identical receptor complex produces opposite fates depending on subcellular localization — a central theme in the pathophysiology of ischemic stroke and neurodegenerative disease.

Pathway context

GluN1/NMDAR sits at the heart of the glutamatergic excitatory signaling pathway: (1) presynaptic glutamate release + ambient glycine/D-serine occupy the GluN2 and GluN1 sites; (2) coincident postsynaptic depolarization (often via AMPA receptors) relieves Mg^{2+} block; (3) channel opening admits Ca^{2+} at the postsynaptic density; (4) Ca^{2+} activates downstream effectors (CaMKII, calcineurin, Ras-ERK, CREB) that execute LTP/LTD, receptor trafficking, and transcriptional programs underlying memory and development. GluN1's own contribution to pathway specificity is the **co-agonist gate** (glycine/D-serine requirement) and the **obligatory scaffolding** that determines whether any NMDAR exists at all.

Why GluN1 is essential and disease-associated

Because GluN1 is the obligatory subunit, its complete loss abolishes all NMDA signaling and is lethal (PMID: 8060614), while point mutations that subtly shift agonist potency, Mg^{2+} block, or gating (e.g., GluN1-M641I; pre-M1 linker variants) cause severe encephalopathy without abolishing the receptor (PMID: 34227748, PMID: 37000222, PMID: 28389307). The genetics bracket GluN1's function from both the null and the dysfunction directions.

Evidence Base

PMID	Study (abbrev.)	Type	How it supports the findings
8060614	*Targeted disruption of NMDA receptor 1 gene...*	In vivo genetics (mouse KO)	GluN1 obligatory & non-redundant; loss abolishes NMDA Ca^{2+}/currents; neonatal lethality (F001)
19794189	*Behavioral deficits... mutant NMDA receptors...*	Mouse hypomorph / review	GluN1 obligatory, no functional homolog, ubiquitous expression; coincidence detection → LTP/ learning (F001)
12805203	*Crystal structures of the NMDA receptor NR1 LBD core*	X-ray crystallography	GluN1 binds glycine/D-serine; clamshell closes on agonist, opens on antagonist (F002, F010)
16281028	*Subunit arrangement and function in NMDA receptors*	Structure/function	GluN1–GluN2A heterodimer is the functional unit (F002)
25008524	*NMDA receptor structures reveal subunit arrangement...*	X-ray (intact receptor)	1-2-1-2 architecture; Hebbian coincidence detection; Mg^{2+} block (F003)
11640925	*An ER retention signal explains surface expression...*	Cell biology	ER retention signal in GluN1 C1 cassette; overridden by PDZ/ phospho (F004)
19620977	*SAP97 and CASK mediate sorting of NMDARs...*	Cell biology	Dedicated dendritic secretory pathway for NMDARs (F004)
18836292	*Role of the fourth membrane domain of NR2B...*	Cell biology	M3 ER-retention determinants ensure only assembled complexes surface (F004)
28389307	*De novo GRIN1 mutations...encephalopathy*	Human genetics / review	Defines GRIN1-related neurodevelopmental disorder phenotype (F005)
34227748	*Recurrent seizure-related GRIN1 variant...*	Functional electrophysiology	GluN1-M641I gain-of-function: ↑ agonist potency, ↓ Mg^{2+} block (F005)
37000222			

PMID	Study (abbrev.)	Type	How it supports the findings
	S1-M1 linker...controls channel opening	Functional variant screen	Pre-M1 linker variants in GRIN1 alter gating; intolerant to variation (F005)
30217972	*A conserved glycine...slow deactivation, high Ca^{2+}*	Electrophysiology/ MD	M4 glycine gating hinge; disease mutations alter gating (F005)
42665662	*Gating/allosteric modulation of GluN1-GluN3A...*	Cryo-EM / physiology	GluN1/GluN3 gated by glycine alone; diheteromeric assembly (F006)
41472680	*Interneuron subsets...GluN1/ GluN3A eGlyR...*	Circuit physiology	eGlyR physiological role in hippocampal networks (F006)
32070387	*Tripartite signalling by NMDA receptors*	Review	Quadruply-liganded (2 glutamate + 2 glycine) state drives gating (F007)
21883211	*A steroid modulatory domain in NR2A...NR1 exon-5...*	Electrophysiology	GluN1 exon-5 tunes proton/ pregnenolone-sulfate modulation (F007)
10844014	*PKC regulates Ca^{2+}-dependent inactivation of NMDARs*	Electrophysiology	Ca^{2+}/CaM inactivation via displacement from GluN1; PKC phosphorylation (F008)
9089692	*Ca^{2+}-dependent inactivation...CaM binding of NR1*	Biochemistry/ physiology	Maps CaM sites to GluN1 C-terminus incl. C1 cassette (F008)
31924235	*microRNA signature of toxic extrasynaptic NMDAR...*	Molecular/cell biology	Synaptic=survival vs extrasynaptic=excitotoxicity dichotomy (F009)
34958875	*NMDAR-mediated preconditioning...iPSC organoids*	Human iPSC model	Extrasynaptic NMDARs drive excitotoxic death in human neurons (F009)
15996549	*Mechanism of partial agonist action at NR1*	Crystallography	GluN1↔GluR2 evolutionary relationship; partial-agonist paradigm (F010)

PMID	Study (abbrev.)	Type	How it supports the findings
15597199	*Evolutionary trace analysis of iGluR sequences*	Bioinformatics/MD	Conserved ligand-binding residues; domain closure essential for gating (F010)
40490524	*NMDAR pathways in ischemic stroke*	Review	Synaptic vs extrasynaptic NMDAR signaling in stroke (context for F009)

Supporting/contextual literature also reviewed includes the role of D-serine and serine racemase as the endogenous co-agonist source (PMID: 37833512, PMID: 36535077); GlyT1 control of ambient glycine and co-agonist site occupancy (PMID: 39059742, PMID: 39770104); subunit-selective allosteric modulators acting via the GluN1 co-agonist site (PMID: 41335128); triheteromeric gating/ketamine sensitivity (PMID: 39954679); and GluN2-subunit diversity governing plasticity and disease (PMID: 31508206, PMID: 39596430). None of these contradict the findings; they reinforce GluN1's identity as the glycine-site-bearing obligatory subunit.

Consistency assessment: The evidence base is unusually convergent. Genetic ablation, human disease variants, multiple independent crystal and cryo-EM structures, electrophysiology, cell-biological trafficking studies, and evolutionary/bioinformatic analyses all point to the same core conclusions, with no material contradictions among the primary sources used for the findings.

Limitations and Knowledge Gaps

1. **Model-system dependence.** Much of the definitive functional and structural work (knockouts, S1S2 crystal structures, Ca^{2+}/CaM inactivation) was performed in rodent or recombinant/*Xenopus* systems. While GluN1 is highly conserved and the human disease genetics align, strictly human experimental data remain comparatively sparse (the iPSC organoid study, PMID: 34958875, is a notable exception).
2. **GluN1 splice-variant diversity.** GluN1 has eight splice variants (combinations of the N-terminal exon-5 and C-terminal C1/C2/C2′ cassettes). This report documents functional consequences of exon-5 and C1, but the full functional map of all variants — their regional expression, trafficking, and modulation — is incomplete.

3. **eGlyR physiology is still emerging.** The GluN1/GluN3 excitatory glycine receptor has only recently been structurally and physiologically characterized. Its precise native stoichiometry, circuit-level roles beyond hippocampal interneurons, and contribution to human disease remain under active investigation.
4. **Mechanistic granularity of the synaptic/extrasynaptic switch.** The dichotomy is well established, but how GluN1-receptor localization is dynamically regulated, and whether GluN1 (versus the GluN2 subtype) directly determines the survival-versus-death signaling, is not fully resolved. Much of the death-signaling specificity has been attributed to the GluN2B C-terminus rather than GluN1.
5. **Quantitative affinity/kinetic parameters** (e.g., exact glycine EC_{50}, Ca^{2+} permeability ratios) vary with subunit composition and were not systematically tabulated here; they depend on the GluN2 partner.
6. **Non-neuronal / peripheral roles.** GluN1 expression and potential functions outside the CNS were not examined.

Proposed Follow-up Experiments / Actions

1. **Comprehensive human splice-variant functional map.** Express all eight human GluN1 splice variants with each GluN2 subtype in a uniform human cellular background (e.g., iPSC-derived neurons) and systematically measure glycine/D-serine potency, proton/polyamine/neurosteroid modulation, surface trafficking, and Ca^{2+}/CaM inactivation — resolving limitation #2.
2. **Isolate the subunit source of the survival/death switch.** Using GluN1 C-terminal mutants (ER-retention, PDZ, CaM, PKC sites) in synaptic- vs extrasynaptic-targeted contexts, dissect whether GluN1's tail contributes to the CREB-on vs CREB-off decision independently of GluN2B — addressing limitation #4.
3. **Structural characterization of GluN1 regulatory states.** Obtain cryo-EM structures of full-length GluN1-containing receptors with the Ca^{2+}/CaM complex bound to the C-terminal tail, to visualize the inactivation mechanism at atomic resolution (extends F008).

4. **Deep mutational scanning of GRIN1.** Systematically assay the functional effect of all possible GRIN1 missense variants (agonist potency, Mg^{2+} block, open probability, surface expression) to build a clinical variant-interpretation resource, extending the targeted studies (PMID: 34227748, PMID: 37000222).
5. **In vivo eGlyR physiology.** Use GluN3-conditional and GluN1-reporter mouse lines (and human tissue where feasible) to define the circuit and behavioral roles of GluN1/GluN3 excitatory glycine receptors (extends F006).
6. **Co-agonist supply in human disease.** Quantify how serine racemase / D-serine and GlyT1 set GluN1 co-agonist site occupancy in human neurons, linking the biochemical supply chain (PMID: 37833512, PMID: 39059742) to therapeutic modulation of NMDAR hypo-/hyperfunction.

Conclusion

GRIN1 encodes GluN1, the obligatory, non-redundant, glycine/D-serine-binding subunit of the NMDA receptor. Its primary molecular function is to supply the co-agonist binding site and co-assemble (two GluN1 + two GluN2, 1-2-1-2) into a ligand- and voltage-gated, Ca^{2+}-permeable cation channel that acts as a Hebbian coincidence detector at the postsynaptic membrane of central neurons. The channel opens only when glutamate (on GluN2), glycine (on GluN1), and depolarization (relieving Mg^{2+} block) coincide, generating the Ca^{2+} signals that drive synaptic plasticity, learning, memory, and neurodevelopment. GluN1 additionally forms glycine-only-gated eGlyRs with GluN3, carries intrinsic regulatory modules (exon-5 allosteric tuning, C1 ER-retention trafficking control, C-terminal Ca^{2+}/CaM feedback inactivation), and mediates opposite survival-versus-death outcomes depending on synaptic versus extrasynaptic localization. Its essential role is confirmed by neonatal-lethal knockouts and severe de novo human encephalopathy variants, and its mechanism by crystal/cryo-EM structures and evolutionary analysis.
